

1M1B Green Internship

FINAL PROJECT SUBMISSION TEMPLATE

(1M1B Green Internship, Batch 6)

1. Basic Details

Intern Name:

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College/University:

Amrita Vishwa Vidyapeetham

Course/Year:

B.Tech – Computer Science and Engineering | 3rd Year

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Brief Bio:

I am a Computer Science undergraduate with a strong interest in applying technology to solve real-world sustainability challenges. My academic journey has helped me build skills in data analysis, visualization, and problem-solving, while projects and internships have strengthened my ability to work on impact-driven initiatives. I enjoy exploring interdisciplinary areas such as environmental sustainability, urban planning, data science, and AI. Through the 1M1B Green Internship, I worked on mapping urban emission hotspots using data-driven methods, which enhanced my understanding of carbon footprint analysis, GIS tools, and community-focused solutions. I am motivated to continue building solutions that combine technology with social and environmental responsibility.

2. Project Overview

Project Title:

Urban Emission Hotspot Mapping for Sustainable City Planning

Project Theme:

Carbon Footprint

Group/Solo:

Group

Group Name:

AGNI

Group Members:

Kuchipudi Yokshith Sai Ram Chowdary - 1M1BGI060098

Dasu Swapnika - 1M1BGI060088

K Geetha Uma Maheswari - 1M1BGI060133

Revanth Kumar Merugu - 1M1BGI060124

Problem Statement:

Urban areas experience high carbon emissions due to dense traffic, industrial activity, and energy consumption. However, these emission hotspots are often not clearly identified or visualized, making it difficult for policymakers and communities to take targeted mitigation actions.

Tools, Frameworks, and Software Used:

Python, Pandas, NumPy, Folium, Tableau, Google Maps data, Open-source datasets, Microsoft Excel

Summary of Work Done :

In this project, I created a grid-based urban emission hotspot map for the Greater Hyderabad Municipal Corporation (GHMC) area. I first collected and cleaned multiple datasets – MMTS transit stops as a proxy for traffic intensity, locations of major industrial areas, the GHMC boundary, and air quality time-series data for Hyderabad. Using Python and GeoPandas, I

divided the city into 600+ grid cells, calculated a traffic score and an industrial score for each cell, and combined them into a normalized emission index between 0 and 1. I then used Folium to build an interactive choropleth heatmap showing emission hotspots, and Tableau Public to create a dashboard with a bar chart of top hotspot areas, a scatter plot of traffic vs industrial drivers, and an AQ trend line. Reverse geocoding was used to assign locality names (e.g., Adikmet, Ameerpet, Begumpet). Finally, I developed targeted recommendations for traffic-dominated, industrial-dominated, and mixed hotspots, and outlined a roadmap for hotspot-based EV deployment, public transport improvements, and industrial emission controls in Hyderabad.

3. Links to Project Deliverables

(All your files and folders should be uploaded to the **Master Google Drive Folder renamed as 1M1B Green Internship Final Projects - Batch 6** in your personal google drive and public links provided below)

SNO	Item	Google Drive Link
1	Master Google Drive Link	[https://drive.google.com/drive/folders/1aZFWWNfiMBy41fMYz268ayzjeswQE62U?usp=sharing]
2	Final PPT (PPT or PDF)	[https://drive.google.com/drive/folders/1u7mBVok0sSAIRXnH2uzi0NWlmuUAHJUG?usp=sharing]
3	Final Project Report / Concept Note(Word Document)	[https://drive.google.com/drive/folders/1BOMaOQtIHSqQnQcnbO5GBp0yiEZrAnlg?usp=sharing]
4	Tableau Dashboard (Direct link or link to the dashboard screenshot uploaded on google drive)	[https://drive.google.com/drive/folders/1xP4bkoZYKiCKfXbM2hkl9Hu0FSvwc-VS?usp=sharing]
5	Project Photos (All the relevant photos of stakeholders interaction, implementation uploaded on a google drive folder)	[https://drive.google.com/drive/folders/1v2DmSBjhdQnfutf]

		qfK6GuqjYs2zBB6cf?usp=sharing
6	Project Videos (Project Video compiled, uploaded on the google drive folder)	[https://drive.google.com/drive/folders/1ykdjYDWBahnN0mmlr7nU07FRiInkVbtC?usp=sharing]
7	Calculate the the final count of your LIP(Create a word document with a summary of your LIP engagements)	[https://drive.google.com/drive/folders/1V_T6a4bjg_7Pu2wfJqdxDZ_0_cydACsN?usp=sharing]
8	Any other supporting document (Raw data, surveys conducted etc)	[https://drive.google.com/drive/folders/1PX_dyP2nISKUi5o9FO_hruggH41tm5A8?usp=sharing]

4. Impact Created

Quantitative Impact:

- 600+ grid cells mapped across GHMC with an emission index combining traffic and industrial influence.
- 6 major high-emission hotspot clusters identified (Adikmet, R.K. Nagar–Malkajgiri, Begumpet–HPS, Ameerpet corridor, Somajiguda–Necklace Road, Gayatri Nagar–Ligamally).
- 3 key Tableau views built (top-hotspot bar chart, traffic vs industrial scatter, AQ trend line) for decision support.
- 5+ stakeholders engaged (faculty mentor, peers, local residents, online viewers via LinkedIn/LIP posts).
- 1 interactive Folium map and 1 public Tableau dashboard created for future use by students and local stakeholders.

Qualitative Impact:

- Created clear visibility of where emissions are concentrated in Hyderabad instead of only city-average pollution numbers.
- Increased awareness among peers and local stakeholders about the need for hotspot-based planning and targeted interventions.
- Built a practical example of how open data and simple modeling can support GHMC's climate and air-quality planning.
- Strengthened my own confidence in using data, maps, and dashboards to tell a compelling environmental story.

5. Learnings from the Internship

Green skills: I learned how to think in terms of urban sustainability, emission hotspots, and targeted mitigation instead of generic “pollution is high” statements.

Data skills: I improved my Python data cleaning, geospatial analysis with GeoPandas, interactive mapping with Folium, and dashboard design with Tableau. I also learned how to integrate multiple datasets (traffic, industrial, AQ) into one model.

Professional skills: I practiced scoping a project, planning timelines, presenting results visually, taking stakeholder feedback, and communicating complex results in simple language.

6. Challenges Faced

- Difficulty in gathering consistent, open data for traffic and industrial emissions in Hyderabad; solved by using proxies like MMTS stops and industrial area locations.
- challenges in working with geospatial libraries and folium/GeoPandas, which took time to debug but improved my understanding.
- Learning Tableau from scratch and connecting multiple CSVs without over-complicating relationships.

7. Acknowledgments

I would like to thank the 1M1B Green Internship team for the structured guidance, my faculty mentor [Name] for feedback on the methodology, my peers who reviewed the dashboard and PPT, and the local residents near hotspot areas who shared their experience of traffic and pollution. I also acknowledge open-data providers (Kaggle, OpenStreetMap, government AQ data) whose datasets made this project possible.

8. Future Roadmap

Timeline	Planned Action (Write your responses below)
Next 1 month	<i>Present the project to my college department and share the dashboard and report with interested faculty and peers; refine the model based on their feedback.</i>
Next 2 months	<i>Reach out to GHMC / local environmental NGOs to share the hotspot map and explore collaboration for a small pilot (e.g., sensor deployment or awareness session in one hotspot area)</i>
Next 6 months	<i>Extend the model with better traffic data (e.g., Google mobility / road counts) and more detailed industrial information; improve CO₂ estimation; publish an updated dashboard and possibly a blog/post</i>
Next 12 months	<i>Replicate the hotspot mapping framework for another Indian city (e.g., Pune or Bangalore) and build a comparative “multi-city emission hotspots” dashboard, while mentoring juniors who want to use this as a starting point.</i>

9. Final Notes / Comments:

This internship helped me connect my interest in data and maps with real climate problems in my own city. It showed me that even a student-level project can create a useful tool for decision makers and communities. I

would love to see 1M1B continue supporting data-driven environmental projects and maybe host a follow-up cohort focused only on city-scale emission hotspot mapping.

